Evaluating Player Performance to Support Team Decisions: St. Louis Cardinals

Marcela Isabel Redondo Hernandez

College of Management, Park University

CIS627HOS2P2026 Sports Analytics

Sahand Hajifar

May 10, 2026

Abstract

This project evaluates player performance for the St. Louis Cardinals by comparing traditional and advanced offensive metrics. The objective is to identify players who may be undervalued or overvalued in lineup decisions and to determine whether on-base plus slugging (OPS) provides a broader evaluation of offensive contribution than batting average (AVG). The dataset includes player statistics from the 2024 and 2025 seasons, including AVG, on-base percentage (OBP), slugging percentage (SLG), OPS, and plate appearances (PA). A diagnostic analytical approach was used, and a custom metric called the Gap was developed to compare player performance with playing time. The results show that performance and playing time are not always aligned, with some high-performing players receiving limited opportunities while others receive more playing time despite lower performance. These findings suggest that OPS provides a broader evaluation of offensive contribution than batting average alone. The analysis supports the recommendation that teams should incorporate OPS into lineup evaluation and align playing time more closely with actual performance. Future improvements could include standardized metrics such as z-scores to improve comparability between variables.

Keywords: sports analytics, OPS, player evaluation, lineup decisions, Major League Baseball, St. Louis Cardinals

Evaluating Player Performance to Support Team Decisions: St. Louis Cardinals

The St. Louis Cardinals are a Major League Baseball team that competes in the National League and operates in an increasingly data-driven environment. Like many professional sports organizations, the Cardinals rely on analytics to support roster construction, lineup decisions, and long-term strategy. The team is currently in a transition phase that emphasizes the development of younger players following the departure of several veteran contributors. Because of this transition, accurately evaluating player performance has become especially important.

Traditional offensive statistics such as batting average have historically been used to evaluate baseball players; however, these metrics do not fully capture a player's overall offensive contribution. Advanced statistics provide a broader perspective by incorporating additional aspects of performance beyond hitting consistency alone. As analytics continues to play a larger role in professional baseball, organizations increasingly use advanced metrics to support more informed and objective decision-making.

This project examines how traditional and advanced offensive metrics can influence lineup evaluation for the St. Louis Cardinals. Specifically, it explores whether OPS provides a broader evaluation of offensive contribution than batting average and whether differences exist between player performance and playing time. By analyzing offensive production alongside usage, this study aims to identify players who may be undervalued or overvalued in lineup decisions.

Research Questions

This project is guided by the following research question:

Which St. Louis Cardinals players may be undervalued or overvalued in lineup decisions when comparing batting average to OPS, and how can OPS provide a broader evaluation of offensive contribution relative to playing time?

Supporting questions include:

1. Which players have a high OPS but relatively low playing time and may be undervalued in lineup decisions?
2. Do players with a high batting average always provide a higher overall offensive contribution?
3. Are there players who may be overvalued when using batting average instead of OPS?

Literature Review

Sports analytics has become an important component of decision-making in professional baseball. Traditional statistics such as batting average have historically been used to evaluate offensive performance; however, these metrics do not always capture a player's complete contribution to team success. Endo et al. (2025) explain that offensive performance in Major League Baseball is evaluated using multiple statistics, including batting average, on-base percentage, slugging percentage, and OPS. The authors emphasize that relying on a single traditional metric may not fully reflect a player's overall offensive ability.

Advanced metrics such as OPS provide a broader evaluation because they combine a player's ability to reach base and generate power into a single measure. According to Endo et al. (2025), OPS reflects both offensive consistency and power-hitting contribution. These types of metrics allow teams to evaluate players more objectively and identify patterns that may not be visible through traditional statistics alone.

According to Alamar (2024), analytics supports decision-making in sports by improving the accuracy and objectivity of player evaluation. Similarly, Zhao et al. (2025) describe how sabermetrics and baseball analytics are increasingly used to support lineup construction, player evaluation, game planning, and organizational strategy. The growing use of analytics in Major League Baseball reflects a broader shift toward data-driven decision-making in professional sports. This project builds on these analytical approaches by examining whether OPS provides a more complete evaluation of offensive contribution than batting average and whether player performance aligns with playing time decisions for the St. Louis Cardinals.

Data Description

The dataset used in this project was obtained from publicly available sources, including Major League Baseball (MLB) statistics (Major League Baseball, n.d.) and Baseball Reference (Baseball Reference, n.d.). The data focuses on offensive statistics for St. Louis Cardinals players during the 2024 and 2025 seasons. Information from both seasons was combined into a single dataset to allow comparison of player performance and playing time across different roster situations.

The dataset includes both traditional and advanced offensive metrics, including batting average (AVG), on-base percentage (OBP), slugging percentage (SLG), on-base plus slugging (OPS), and plate appearances (PA). Batting average was included as a traditional measure of offensive performance, while OPS was used as a broader metric because it combines a player's ability to reach base and generate offensive power. Plate appearances were included to represent player usage and playing time in lineup decisions.

The final dataset contains 42 player observations and was organized using Microsoft Excel and Python in Kaggle to support the analysis of offensive contribution and lineup usage.

Results

Data Analysis

The dataset was reviewed to ensure consistency and reliability before the analysis was performed. All variables were converted to the appropriate data types, and player information was checked to ensure consistency across both seasons. No missing values were identified in the final dataset, resulting in 42 complete player observations available for analysis.

Summary statistics revealed variation in both offensive performance and playing time across players. Batting average remained relatively consistent throughout the dataset, while OPS showed greater variation, indicating differences in overall offensive contribution. Plate appearances also varied considerably, reflecting differences in lineup usage and player opportunities during the 2024 and 2025 seasons.

Figures 1 through 3 provide an overview of the dataset, its structure, and the summary statistics used to support the analysis. Together, these figures confirm that the dataset is complete, consistent, and suitable for evaluating player performance and lineup usage patterns.

Figure 1

Dataset preview showing player performance metrics

	Player	AVG	PA	OBP	SLG	OPS	Season
0	Pedro Pagés	0.238	218	0.281	0.376	0.657	2024
1	Paul Goldschmidt	0.245	654	0.302	0.414	0.716	2024
2	Nolan Gorman	0.203	402	0.271	0.400	0.671	2024
3	Masyn Winn	0.267	637	0.314	0.416	0.730	2024
4	Nolan Arenado	0.272	635	0.325	0.394	0.719	2024

Figure 2

Dataset structure and data types confirming data quality

```
<class 'pandas.core.frame.DataFrame'>
Index: 42 entries, 0 to 21
Data columns (total 7 columns):
 #   Column  Non-Null Count  Dtype  
---  ------  --------------  -----  
 0   Player  42 non-null     object 
 1   AVG     42 non-null     float64
 2   PA      42 non-null     int64  
 3   OBP     42 non-null     float64
 4   SLG     42 non-null     float64
 5   OPS     42 non-null     float64
 6   Season  42 non-null     int64  
dtypes: float64(4), int64(2), object(1)
memory usage: 2.6+ KB
```

Figure 3

Summary statistics highlighting variation in performance and playing time

	AVG	PA	OBP	SLG	OPS	Season
count	42.000000	42.000000	42.000000	42.000000	42.000000	42.000000
mean	0.220833	290.071429	0.288643	0.341000	0.629500	2024.523810
std	0.051461	221.013342	0.059750	0.091473	0.143297	0.505487
min	0.077000	6.000000	0.133000	0.077000	0.271000	2024.000000
25%	0.198750	71.750000	0.262250	0.283500	0.552250	2024.000000
50%	0.232500	277.000000	0.292500	0.367000	0.661500	2025.000000
75%	0.257750	460.250000	0.325000	0.415500	0.727250	2025.000000
max	0.301000	654.000000	0.380000	0.468000	0.848000	2025.000000

Analysis Approach

This project uses diagnostic analysis to examine how player performance relates to playing time, which is a common approach in sports analytics decision-making (Alamar, 2024). OPS was used to represent offensive contribution, while plate appearances (PA) were used to represent player usage in lineup decisions.

To connect these two aspects, a custom metric called the Gap was created. The Gap compares a player's OPS rank with their plate appearance rank, making it easier to identify situations in which performance and usage do not align. This approach helps highlight players who are performing well offensively but are receiving fewer opportunities, as well as players who receive greater playing time despite lower offensive performance.

This analytical approach focuses on identifying potential inefficiencies between player performance and lineup usage in a way that can support more informed decision-making. However, one limitation of the analysis is that it only considers offensive performance and does not include additional factors such as defensive contribution, injuries, coaching strategy, or specific game situations. In addition, the analysis is limited to two seasons of data and may not fully capture long-term performance trends.

Another limitation is that OPS and plate appearances operate on different scales, and ranking players may compress differences between observations. Future improvements could include the use of standardized metrics such as z-scores to improve comparability between variables and preserve magnitude differences across player performance.

Performance Analysis

The analysis was completed using Python in a Kaggle notebook, where the datasets from the 2024 and 2025 seasons were cleaned, prepared, and combined for analysis. Players were compared using OPS to represent offensive performance and plate appearances (PA) to represent playing time and lineup usage.

Using the Gap metric, players were ranked to evaluate how offensive contribution aligns with playing time. This approach made it possible to identify differences between players who contribute offensively and how frequently they appear in lineup decisions. Some players demonstrated strong OPS with fewer plate appearances, suggesting potential undervaluation, while others received greater playing time despite lower offensive performance.

The following figures illustrate these patterns and provide a clearer understanding of how player performance and usage relate across the dataset.

Figure 4

Gap metric calculation comparing OPS rank and plate appearance rank

	Player	AVG	PA	OBP	SLG	OPS	Season	OPS_rank	PA_rank	Gap
0	Pedro Pagés	0.238	218	0.281	0.376	0.657	2024	22.0	23.0	-1.0
1	Paul Goldschmidt	0.245	654	0.302	0.414	0.716	2024	13.0	1.0	12.0
2	Nolan Gorman	0.203	402	0.271	0.400	0.671	2024	19.0	15.5	3.5
3	Masyn Winn	0.267	637	0.314	0.416	0.730	2024	11.0	3.0	8.0
4	Nolan Arenado	0.272	635	0.325	0.394	0.719	2024	12.0	4.0	8.0

The Gap metric was used to compare player performance (OPS) with playing time (PA). This table shows how players were ranked based on OPS and plate appearances to create the Gap metric. It provides a clear comparison between offensive contribution and usage and helps identify where those two factors do not align.

Figure 5

Undervalued players with strong offensive performance but limited playing time

	Player	AVG	PA	OBP	SLG	OPS	Season	OPS_rank	PA_rank	Gap
13	José Fermín	0.283	70	0.377	0.417	0.793	2025	5.0	32.0	-27.0
16	Luken Baker	0.235	41	0.366	0.324	0.689	2025	14.0	37.0	-23.0
19	Luken Baker	0.175	49	0.286	0.400	0.686	2024	16.0	35.0	-19.0
10	Iván Herrera	0.301	259	0.372	0.428	0.800	2024	4.0	22.0	-18.0
9	Willson Contreras	0.262	358	0.380	0.468	0.848	2024	1.0	19.0	-18.0

This figure highlights players who demonstrate strong OPS but receive fewer plate appearances. These results suggest that their offensive performance is not fully reflected in lineup usage, indicating potential undervaluation in playing time decisions. These findings may indicate opportunities to optimize lineup decisions during player development and roster transitions.

Figure 6

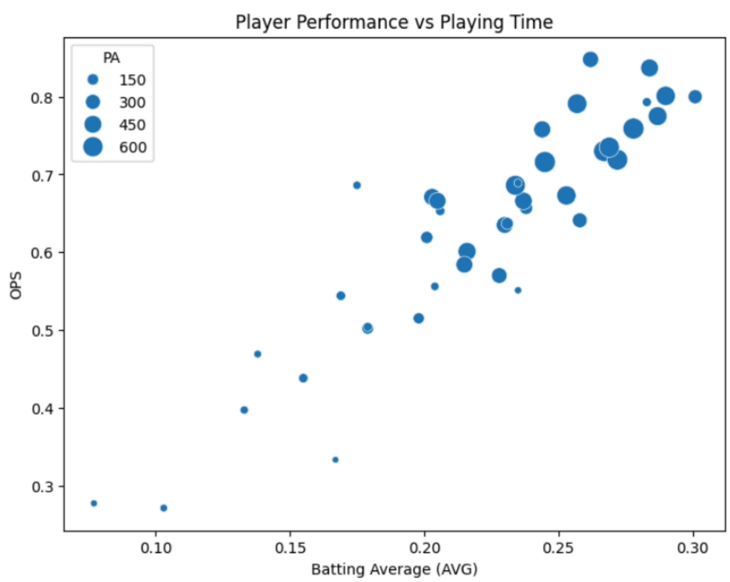
Overvalued players with higher playing time relative to offensive performance

	Player	AVG	PA	OBP	SLG	OPS	Season	OPS_rank	PA_rank	Gap
6	Victor Scott II	0.216	463	0.305	0.296	0.601	2025	28.0	11.0	17.0
1	Paul Goldschmidt	0.245	654	0.302	0.414	0.716	2024	13.0	1.0	12.0
7	Jordan Walker	0.215	396	0.278	0.306	0.584	2025	29.0	17.0	12.0
5	Lars Nootbaar	0.234	583	0.325	0.361	0.686	2025	16.0	6.0	10.0
6	Michael Siani	0.228	334	0.285	0.285	0.570	2024	30.0	20.0	10.0

This figure highlights players who receive greater playing time despite lower OPS values. The results suggest that some players may receive more opportunities than their offensive performance alone would support, indicating that lineup usage may not always align with overall offensive contribution.

Figure 7

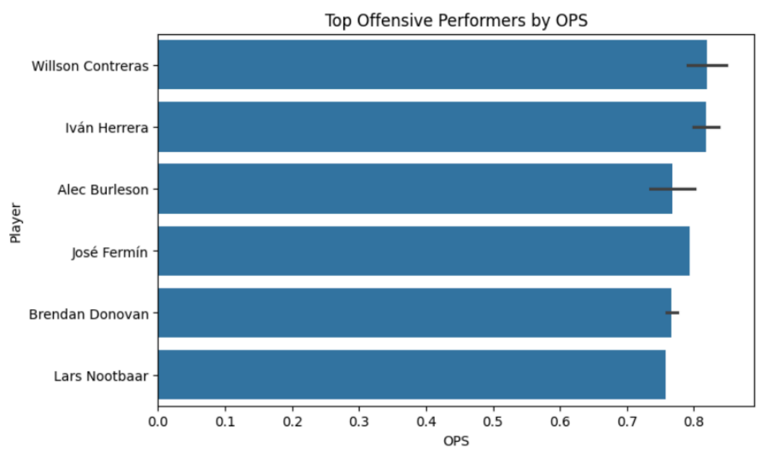
Scatter plot showing the relationship between batting average, OPS, and playing time



This scatter plot illustrates how offensive performance and playing time relate across players. Some players with higher OPS appear as smaller markers, indicating limited playing time, while others with lower OPS appear as larger markers. These patterns suggest that offensive contribution and lineup usage are not always aligned. The visual distribution of players also suggests that OPS captures broader differences in offensive performance than batting average alone.

Figure 8

Top offensive performers based on OPS



This bar chart highlights the players with the strongest offensive performance based on OPS. The figure provides a direct comparison across players and demonstrates how OPS can help identify players contributing the most offensively.

Together, these results show that player performance and playing time do not always align. The findings suggest that OPS provides a broader evaluation of offensive contribution than batting average alone because it captures both on-base ability and offensive power. The analysis also demonstrates how analytics can support more effective lineup and roster decisions by identifying players whose contributions may not be fully reflected through traditional statistics alone.

Conclusion

This project examined how offensive performance and playing time relate for St. Louis Cardinals players during the 2024 and 2025 seasons. The objective was to evaluate whether OPS provides a broader evaluation of offensive contribution than batting average and to identify players who may be undervalued or overvalued in lineup decisions based on offensive performance and usage.

The analysis demonstrated that OPS provides a broader evaluation of offensive contribution because it considers both a player's ability to reach base and generate power. Batting average alone does not fully capture a player's overall offensive value. The results also showed that player performance and playing time were not always aligned. Some players demonstrated strong offensive performance but received fewer opportunities, while others received greater playing time despite lower OPS values.

The Gap metric helped identify differences between offensive contribution and lineup usage. The scatter plot and bar chart supported these findings by showing variation between OPS, batting average, and playing time across players. The results suggest that analytics can support more effective lineup and roster decisions by helping organizations identify players whose contributions may not be fully reflected through traditional statistics alone.

One limitation of this analysis is that OPS and plate appearances operate on different scales, while ranking players may compress differences between observations. Future research could improve this methodology by using standardized metrics such as z-scores to improve comparability and preserve magnitude differences across player performance. Additional extensions could include defensive statistics, injury history, situational performance, and larger multi-season datasets to provide a more comprehensive evaluation of player value.

These results are especially relevant during a transition period in which organizations must evaluate younger players and allocate opportunities effectively. The growing importance of analytics in modern baseball decision-making also demonstrates how advanced metrics can provide insights beyond traditional statistics when evaluating player performance for the St. Louis Cardinals.

References

Alamar, B. C. (2024). *Sports analytics: A guide for coaches, managers, and other decision makers*. Columbia University Press.

Baseball Reference. (n.d.). *St. Louis Cardinals statistics (2024).* https://www.baseball-reference.com/teams/STL/2024.shtml

Baseball Reference. (n.d.). *St. Louis Cardinals statistics (2025).* https://www.baseball-reference.com/teams/STL/2025.shtml

Endo, N., Yamaguchi, Y., Uchino, H., & Hashimoto, C. (2025). Evaluation of Major League Baseball offensive statistics underscores Shohei Ohtani's exceptional batting performance. *IAENG International Journal of Applied Mathematics,* 55(7), 1921–1925 https://www.iaeng.org/IJAM/issues_v55/issue_7/IJAM_55_7_02.pdf

Major League Baseball. (n.d.). *St. Louis Cardinals player statistics.* https://www.mlb.com/cardinals/stats/

Zhao, W., Akella, V. S., Yang, S., & Luo, X. (2025). Machine learning in baseball analytics: Sabermetrics and beyond. *Information, 16*(5), 361. https://www.mdpi.com/2078-2489/16/5/361